

HEATHER PRINCE

Postdoctoral Research Associate
Department of Physics and Astronomy, Rutgers University
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EDUCATION

- 2022 **PhD Astrophysical Sciences** Princeton University, Princeton NJ
2016 **MSc Applied Mathematics** cum laude, University of KwaZulu-Natal, South Africa
2013 **BSc (Honors) Applied Mathematics** summa cum laude, University of KwaZulu-Natal
2012 **BSc Mathematics and Physics** with distinction, Rhodes University, South Africa

EMPLOYMENT

- 2022- **Postdoctoral Research Associate** Department of Physics and Astronomy, Rutgers University
Co-lead of the Dark Energy Science Collaboration's Forecasting Topical Team

TEACHING EXPERIENCE

Rutgers University, Piscataway NJ

- 2024 Instructor of Record, Principles of Astrophysics II

Rutgers University New Jersey Scholarship and Transformative Education in Prisons (NJSTEP)

- 2023 Instructor, Mathematics in the Courtroom, Northern State Prison

Raritan Valley Community College Returning and Incarcerated Student Education Program (RISE)

- 2023 Instructor of Record, Elementary Algebra, East Jersey State Prison

Princeton University Prison Teaching Initiative (Raritan Valley Community College courses taught by a team of 4 instructors)

- 2025 Instructor, Introduction to Astronomy, Northern State Prison

- 2024 Assistant Instructor, Introduction to Astronomy, East Jersey State Prison

- 2023 Curriculum Developer and Lead Instructor, Introduction to Astronomy, Edna Mahan Correctional Facility

- 2022 Lead Instructor, Basic Mathematics, East Jersey State Prison

- 2020 Instructor, General Physics I (with lab), East Jersey State Prison

- 2019 Instructor, Basic Mathematics

- 2018 Assistant Instructor, Pre-Calculus, East Jersey State Prison

Princeton University

- 2017 Assistant in Instruction, Planets in the Universe

University of KwaZulu-Natal, South Africa

- 2014-2015 Physics Tutor

Rhodes University, South Africa

- 2012 Physics Laboratory Demonstrator

- 2011-2012 Physics Academic Development Program Tutor

MENTORSHIP

- 2024 **Advisor** for month-long project in research methods class, Rutgers University
2023-2024 **Research Advisor** for Senior Honors Thesis, Rutgers University
2018-2020 **Mentor** for Women in Physics Mentorship Program, Princeton University

OUTREACH

- 2024 Skype a Scientist virtual talk and Q&A, Brooklyn Public Library
 Illuminating Dark Energy Using Galaxy Observations, Amateur Astronomers Society NJ
2023 STEM Information Session, East Jersey State Prison
2022 Infrared light and the James Webb Space Telescope, elementary school science day
2019 How Big Is Space? Skype a Scientist virtual elementary school talk
 Junior Girl Scout Career Talk
 Girl Scout Career Day Panel
2015 Careers in Astrophysics presentations at high schools, Durban, South Africa

SCIENTIFIC PRESENTATIONS

- 2025 **Preparing for LSST: 3x2pt forecasts and the cosmology inference pipeline**
 American Astronomical Society Meeting, Washington DC
2024 **Dark Energy Forecasting for the Legacy Survey of Space and Time (LSST)**
 American Astronomical Society Meeting, New Orleans
2023 **Combined Probe Forecasting with DESC Analysis Tools**
 LSST Dark Energy Science Collaboration Meeting, SLAC
 Foreground-marginalized likelihood for constraining primordial gravitational waves with the BICEP and Keck experiments
 American Astronomical Society Meeting, Seattle
2021 **Characterizing primordial fluctuations with the Atacama Cosmology Telescope**
 Columbia University
 Constraining isocurvature fluctuations with the Atacama Cosmology Telescope and the Simons Observatory
 Rutgers University
2020 **Data compression and likelihood-free inference in cosmology**
 American Astronomical Society Meeting, Hawaii
2019 **Data compression in cosmology: a lightweight Planck likelihood**
 Cosmology on Safari Conference, South Africa
2018 **Variability of blazars at 148 GHz in the Atacama Cosmology Telescope data**
 Atacama Cosmology Telescope collaboration meeting, Princeton University
2015 **Cross-correlating CMB lensing with post-reionization HI intensity mapping surveys**
 Square Kilometre Array Bursary Conference, Stellenbosch, South Africa
 Real space lensing reconstruction using cosmic microwave background polarization
 Cosmology on Safari Conference, South Africa
2014 **Real space lensing reconstruction using cosmic microwave background polarization**
 South African Institute of Physics Conference, Johannesburg, South Africa

TRAINING

- 2025 **Increasing student learning and inclusion in your classroom: Strategies from the Faculty Teaching Institute**, American Astronomical Society Meeting Workshop
- 2021 **Fundamentals of Machine Learning**, Princeton University
- 2021 **Modern Statistics**, Princeton University
- 2017 **Physics of the Universe**, Princeton University
- 2016 **Software Engineering for Scientific Computing**, Princeton University

PUBLICATIONS

- 10 W. Coulton, M. Madhavacheril, A. Duivenvoorden, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: High-resolution component-separated maps across one-third of the sky*. Phys. Rev. D Volume 109 Issue 6 (2024). arXiv:2307.01258.
- 9 M. Madhavacheril, F. Qu, B. Sherwin, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: DR6 Gravitational Lensing Map and Cosmological Parameters*. The Astrophysical Journal Volume 962 Issue 2 (2024). arXiv:2304.05203.
- 8 F. Qu, B. Sherwin, M. Madhavacheril et al. (including **H. Prince**). *The Atacama Cosmology Telescope: A Measurement of the DR6 CMB Lensing Power Spectrum and its Implications for Structure Growth*. arXiv e-prints (2023). arXiv:2304.05202.
- 7 **H. Prince** and J. Dunkley, *Compressed Python likelihood for large scale temperature and polarization from Planck*. Phys. Rev. D Volume 105 Issue 2 (2022), arXiv:2104.05715.
- 6 S. Aiola, E. Calabrese, L. Maurin, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: DR4 maps and cosmological parameters*. JCAP 12 (2020) 047. arXiv:2007.07288.
- 5 S. K. Choi, M. Hasselfield, S.-P. P. Ho, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: a measurement of the Cosmic Microwave Background power spectra at 98 and 150 GHz*. JCAP 12 (2020) 045. arXiv:2007.07289.
- 4 **H. Prince** and J. Dunkley. *Data compression in cosmology: A compressed likelihood for Planck data*. Phys. Rev. D Volume 100 Issue 8 (2019), arXiv:1909.05869.
- 3 R. Datta, S. Aiola, S. K. Choi, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: two-season ACTPol extragalactic point sources and their polarization properties*. MNRAS 486, 5239 (2019). arXiv:1811.01854.
- 2 **H. Prince**, K. Moodley, J. Ridd, M. Bucher. *Real space lensing reconstruction using cosmic microwave background polarization*. JCAP 01 (2018) 034., arXiv:1709.02227.
- 1 B. Partridge, L. Bonavera, M. López-Caniego, et. al. (including **H. Prince**). *Can CMB Surveys Help the AGN Community?* Galaxies, 5, 47 (2017).

PREPRINTS

- H. Prince**, E. Calabrese, J. Dunkley. *A foreground-marginalized 'BK-lite' likelihood for the tensor-to-scalar ratio*. arXiv eprint (2024). arXiv:2403.00085.
- K. Moodley, W. Naidoo, **H. Prince**, A. Penin. *A cross-bispectrum estimator for CMB-HI intensity mapping correlations*. arXiv eprint (2023). arXiv:2311.05904.

AWARDS

Princeton University

2016 Dean's Grant (five year funding package)
Princeton University First Year Fellowship

University of KwaZulu-Natal

2014 Vincent Maphai MSc Scholarship
Square Kilometer Array MSc Bursary
2013 National Research Foundation of South Africa BSc (Honors) Scholarship

REFERENCES

Eric Gawiser

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Jill Stockwell

Associate Director, Prison Teaching Initiative
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Jenny Greene

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Jo Dunkley

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